LAB 3: Path Planning

Preparation: Please print out the maps appended to this guide and place them on the table right in front of the robot. You can assume the map is known. So, please make the necessary measurements for the size and location of the obstacles in the map.

You can choose where to place the map in front of the robot. Please be aware that your positioning matters. Suggested placement is right next to the base of the robot at right angles. The long edge should intersect with the robot's base. At the home position the arm should be dividing the map into two halves.

Task 1: Please implement a path planner using **i)** Artificial Potential Fields, **ii)** one of the probabilistic methods (PRM, or RRT) for the first two maps.

Your code should include the helper functions foe readability. A suggested structure is given below. Your implementation should have a <u>similar</u> organization, this is mandatory:

<u>For artificial potential fields:</u>
main()
get_joint_angles() #IK solver
get_distance()
get_repulsion_gradient()

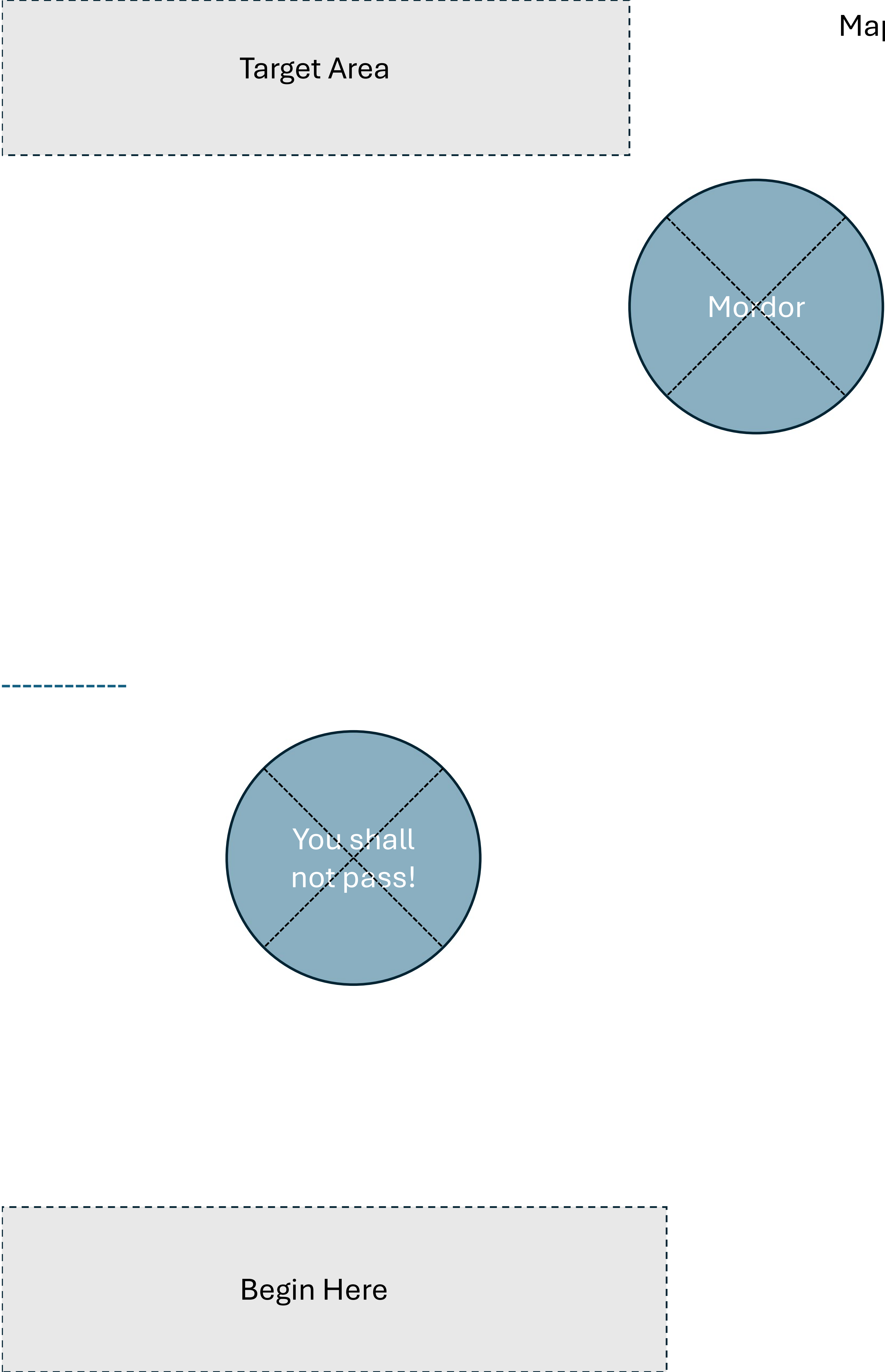
<u>For PRM:</u>
main()
get_joint_angles()
point_collision_check() #in-or-out test
edge_collision_check() #local planner
get_neighbours()
get_graph()
simple_dijkstra()

<u>For RRT:</u>
main()
point_collision_check() #in-or-out test
edge_collision_check() #local planner
node_sampler()
get_nearest_node()
get_path()

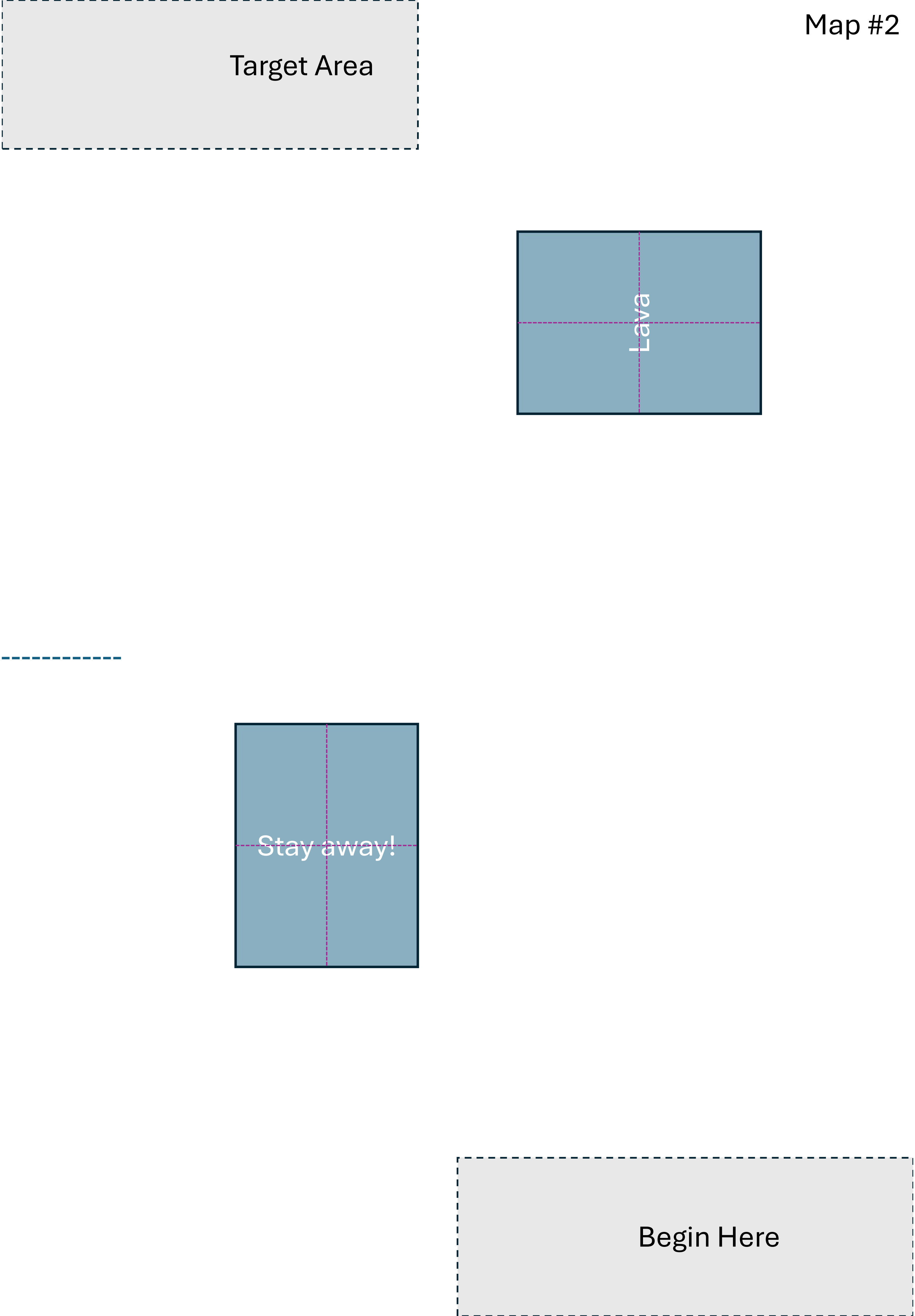
Task 2: Test your planner for 3 different initial and final points. The maps indicate the regions on the map where the initial and final points should belong to. Record your planner's moves as it travels and plot the resultant path on a 2D graph, indicating how it avoids the obstacle. Please briefly comment on its performance, comparing the total travel distance in each case.

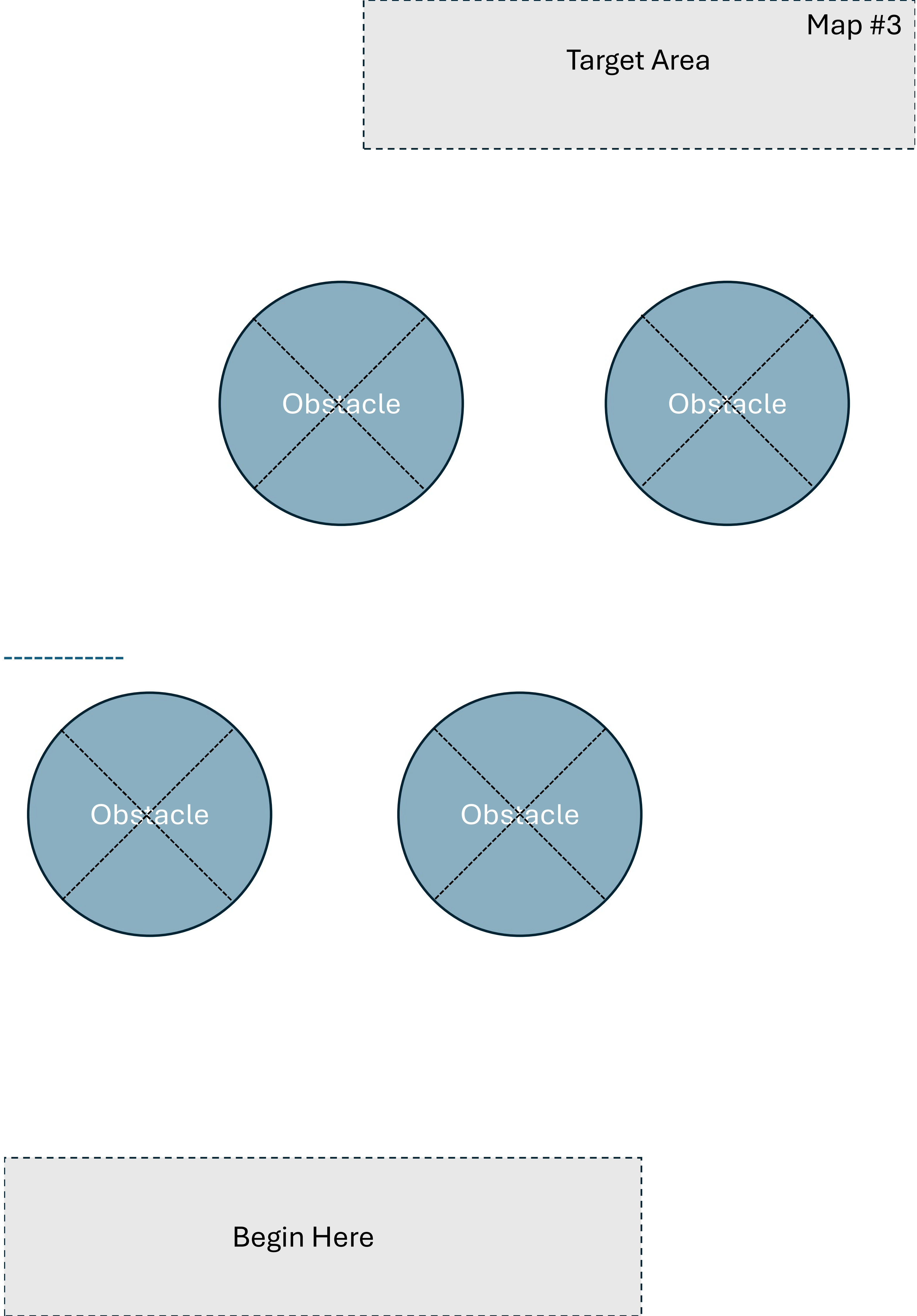
(BONUS) Task 3: Implement a planner method of your choosing on the map #3 and #4. Show the planner's performance by following the instructions in Task 2.

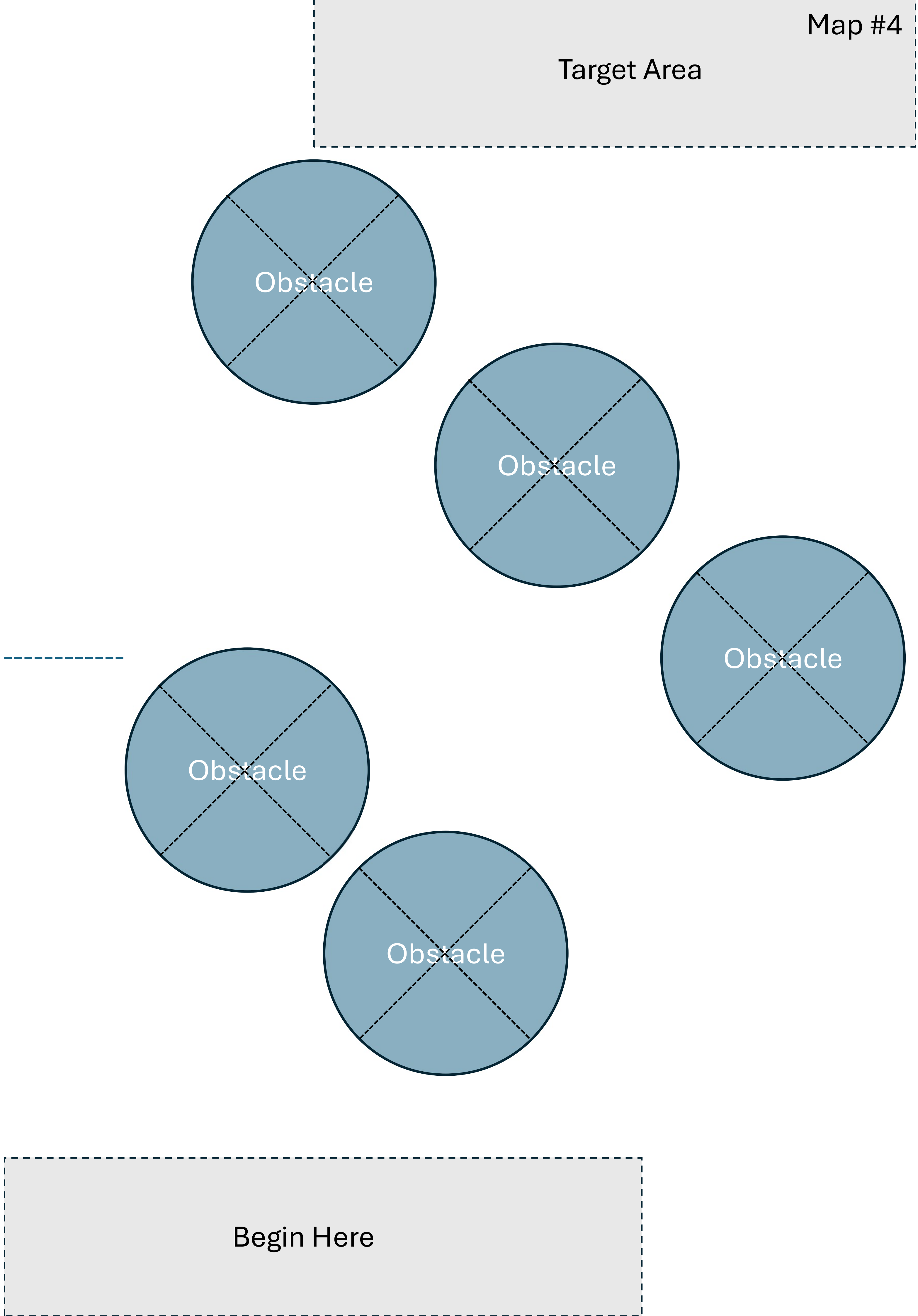
Map #1



Map #2




Map #3
Target Area
Obstacle
Obstacle
Obstacle
Obstacle
Begin Here


Map #4
Target Area
Obstacle
Obstacle
Obstacle
Obstacle
Obstacle
Begin Here